MILITARY SPECIFICATION

FUZE, M509A1, SPOILER SWITCH ASSEMBLY FOR PARTS FOR, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This Military Specification covers parts for assembling and packing of Fuze, M509A1, Spoiler Switch Assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation - Packaging, Methods Of
MIL-B-117	- Bag, Sleeve and Tubing - Interior Packaging
MIL-B-3106	- Board, Composition, Water-resistant, Solid (For Filler or Cushioning Pads)
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-I-45607	- Inspection Equipment, Supply and Maintenance of
MIL-F-50449	- Filler, Sheet Form (For Use In Ammunition Containers)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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FEDERAL

PPP-F-320	- Fiberboard; Corrugated and Solid, Sheet Stock, (Container Grade), and Cut Shapes
PPP-B-601	- Box, Wood, Cleated Plywood
PPP-B-636	- Box, Shipping, Fiberboard

STANDARDS

MILITARY

MIL-STD-1169	- Packaging, Packing And Marking for Shipment of Inert Ammunition Components

DRAWINGS

PICATINNY ARSENAL

PRODUCT AND PACKAGE DRAWINGS

9322034	- Spoiler Switch Assembly, Fuze M509A1
9321944	- Power Supply, Fuze M509
9318154	- Grounding Sleeve Assembly
9283141	- Projectile, 105MM, HEAT-T, M456A1E2, Loading Assembly
9276311	- Cartridge, 105MM, HEAT-T, M456A1E2 Retrofit Assembly

FRANKFORD ARSENAL

10521642	- Spike

3. REQUIREMENTS

3.1 General. -Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Spoiler Switch Assembly.-Spoiler Switch Assemblies shall comply with all requirements specified on dwg. 9322034 and all the requirements specified in applicable specifications.

3.3 Workmanship.-All parts and assemblies shall be free of chips, dirt, grease, rust, burrs, cracks and other foreign material. The cleaning methods used shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

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3.4 Proving Ground.-The applicable loaded projectile shall comply with the following requirements when assembled with a spoiler switch assembly and tested as specified in 4.4.3.1.

3.4.1 Graze Functioning.-The projectile shall detonate "high-order" on initial impact with a gravel pad located 1000 ± 200 meters from the gun muzzle.

3.4.2 Safety.-There shall be no indication of spoiler switch assembly separation or breakup in the gun bore or in flight prior to impact.

3.5 First Article Inspection.-This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections.-The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission.-The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of 30 each of every component and every assembly. (Where a multiple cavity mold is used in the manufacturing process, a minimum of 5 samples shall be submitted from each mold cavity. Each part shall be identified with its mold cavity).

4.3.2 Inspections to be Performed.-The First Article parts submitted in accordance with paragraph 4.3.1 will be subjected by the Government to all examinations and tests specified in 4.4.2 of this specification and to any or all requirements of the applicable drawings. 5 each of the sample

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assemblies (2 each if multiple cavity mold is used) shall be temperature conditioned at 180°F ± 10°F for 24 ± 1 hours and then held at room ambient temperature for 4 hours minimum prior to inspection.

4.3.3 Rejection.-See MIL-A-48078 (PA).

4.4 Quality Conformance Inspection

4.4.1 Inspection Lot Formation.-Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination.-See MIL-A-48078

a. Sampling Plans.-Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48754 (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cup	1 OF 1	9276312
			NEXT HIGHER ASSEMBLY 9276247

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None Defined				
Major					
101	Location of first (from forward) basic diameter		0.40%	3.2	Gage
102	Location of second basic diameter		0.40%	3.2	Gage
103	Location of third basic diameter		0.40%	3.2	Gage
104	Location of aft basic diameter		0.40%	3.2	Gage
105	Depth to forward internal shoulder		0.40%	3.2	Gage
106	Depth to inside nose point		0.40%	3.2	Gage
107	Flange outside diameter		0.40%	3.2	Gage
108	Radius at flange		0.40%	3.2	Gage
109	Inside radius at nose		0.40%	3.2	Gage
110	Wall Thickness (check a 2 locations on flange, 4 through sidewall)		0.40%	3.2	Gage
111	Thickness through nose point		0.40%	3.2	Gage
112	True position of holes (4 sets)		0.40%	3.2	Gage
Minor					
201	Outside angle of nose point		0.65%	3.2	Gage
202	Inside angle of nose point		0.65%	3.2	Gage
203	Angle of flange		0.65%	3.2	Gage
204	Protective finish missing or incomplete		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
4.4.2.2	Impact Switch Assembly		1	2	9276247
					NEXT HIGHER ASSEMBLY 9322034
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major A					
101	Runout of flange		0.40%	3.2	Gage (Note)
102	Aft body runout		0.40%	3.2	Gage (Note)
103	Forward body runout		0.40%	3.2	Gage (Note)
104	Insulation resistance		100%	3.2	4.5.2 (Note)
Major B					
131	Height, min.		0.40%	3.2	Gage
132	Flange outside diameter		0.40%	3.2	Gage
133	Depth to forward internal basic diameter		0.40%	3.2	Gage
134	Location of aft internal basic diameter		0.40%	3.2	Gage
135	Location of middle internal basic diameter		0.40%	3.2	Gage
136	Inside diameter at aft internal shoulder,	max.	0.40%	3.2	Gage
137	Dimension to thread shoulder		0.40%	3.2	Gage
138	Inside diameter forward of threads, max.		0.40%	3.2	Gage
139	Thickness of flange		0.40%	3.2	Gage
140	Thread pitch diameter, max		0.40%	3.2	Gage
141	Thread minor diameter, max.		0.40%	3.2	Gage
142	Dimension to spoiler shoulder		0.40%	3.2	Gage
143	Outside diameter at spoiler shoulder		0.40%	3.2	Gage
144	Wall thickness through spike ribs		0.40%	3.2	Gage
145	Thickness of spike ribs		0.40%	3.2	Gage
146	Thickness of flange ribs		0.40%	3.2	Gage
147	Any feature or operation missing		0.40%	3.2	Visual

NOTES:
Optional reduced inspection (para 4.5.3) not applicable.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.2	Impact Switch Assembly		9276247
			NEXT HIGHER ASSEMBLY 9322034

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major B					
248	Excessive flash between ribs		0.40%	3.2	Visual/Scale
249	Angle of flange outside edge		0.40%	3.2	Gage
Minor					
201	Width of narrow ribs on flange		0.65%	3.2	Gage
202	Width of wide ribs on flange		0.65%	3.2	Gage
203	Diameter of outside circular flange rib		0.65%	3.2	Gage
204	Diameter of inside annular flange rib		0.65%	3.2	Gage
205	Width of inside annular flange rib		0.65%	3.2	Gage
206	Outside diameter of flange sealing groove		0.65%	3.2	Gage
207	Width of flange sealing groove		0.65%	3.2	Gage
208	Depth of sealing groove		0.65%	3.2	Gage
209	Inside radius at flange		0.65%	3.2	Visual
210	Evidence of poor workmanship		0.65%	3.3	Visual
211	Thread pitch diameter, min.		0.65%	3.2	Gage
212	Thread minor diameter, min.		0.65%	3.2	Gage
213	True position of inside diameter at aft internal shoulder		0.65%	3.2	Gage
214	True position of inside diameter forward of threads		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48754 (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Spoiler				9322030
					NEXT HIGHER ASSEMBLY
					9322034
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wall thickness, min.		0.40%	3.2	Gage
Minor					
201	Inside diameter		0.65%	3.2	Gage
202	Outside diameter at flange		0.65%	3.2	Gage
203	Inside diameter at tabs		0.65%	3.2	Gage
204	Any operation or feature missing		0.65%	3.2	Visual
205	Evidence of poor workmanship		0.65%	3.3	Visual
206	Height		0.65%	3.2	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-F-48754 (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Spoiler Switch Assembly, Fuze, M509A1		9322034
			NEXT HIGHER ASSEMBLY
			9276311

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Safety (note)		4.4.3.1.2	3.4	Visual/Photographic
Major					
101	Graze functioning (note)		4.4.3.1.1	3.4	4.5.4.1
102	Spoiler missing or insecurely assembled		0.40%	3.2	Visual/Manual
Minor					
201	Presence of excess adhesive after assembly		0.65%	3.2	Visual
202	Height, max.		100%	3.2	Gage

NOTES:

Proving ground

SARPA-QA FORM 2567 FEB 74 4810

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4.4.3 Testing

4.4.3.1 Proving Ground

4.4.3.1.1 Graze Functioning.-A sample of twenty (20) spoiler switch assemblies shall be randomly selected from each lot for this test. Each sample assembly shall be assembled onto a cartridge comprised of components (projectile, fuze, shaped-charge liner, etc.) previously tested and accepted in accordance with applicable specifications. If six (6) or more projectiles fail to comply with the applicable requirement, the spoiler switch assembly lot shall be rejected. If one (1) or more projectiles fail to initially impact the gravel pad, the spoiler switch assembly lot shall be **referred.**

4.4.3.1.2 Safety.-All cartridges fired in proving ground testing shall be observed for spoiler switch assembly separation or breakup. If any spoiler switch assembly fails to comply with the applicable requirement, the lot shall be rejected.

4.5 Test Method and Procedures

4.5.1 Materials, Components and Processes.-Compliance with all requirements of Section 3 of this specification shall be ascertained by current and continuing examination of inspection and test data to determine that all components (parts, subassemblies, and materials) have been inspected and tested and found to comply with their respective drawing and specification requirements, and that all specified manufacturing processes have been followed.

4.5.1.1 Visual and Mechanical Inspection.-The visual, manual and mechanical inspections and tests shall verify compliance with the requirements of Sections 3 and 5 of this specification in accordance with 4.4.2 herein.

4.5.2 Insulation Resistance.-Minimum insulation resistance shall be tested with the impact switch placed over an electrically conductive mandrel. The mandrel shall have the form and maximum external dimensions of an M456 spike (Dwg. 10521642). The mandrel shall include a Grounding Sleeve (Dwg. 9318154) having maximum height and external dimensions (except threads). The impact switch being tested shall be held down on the mandrel by a 15 + 5 pound load on the impact switch nose end. Insulation resistance shall be tested between the mandrel and the impact switch cup.

4.5.3 Dimensional Control of Cast, Stamped, or Molded Parts.-In place of the normal sampling associated with the classification of Defects, and subject to approval of the

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Government procuring agency, the optional sampling plans in paragraphs 4.5.3.1, 4.5.3.2, and 4.5.3.3 may be used for dimensional control of cast, stamped, or molded parts. Use of these optional plans is limited to those dimensions and features controlled by the tool, die, or mold cavity. For all specified tests, visual examinations, and examinations of dimensions or features not developed exclusively by the tool, die, or mold cavity, these optional plans are not applicable. In the event of in-process rejects(s) during optional sampling, corrective action applicable to parts produced prior to the rejected, and subsequent to the last acceptable sample, shall require approval of the Government procuring agency.

4.5.3.1 Cast Parts.-A sample of at least five (5) parts (as cast) from each cavity shall be dimensionally inspected to qualify a new or reworked cavity for use in production or as a control of the cavity during production. Inspection for control of each cavity during production shall be performed on the above quantities from each cavity after production of 5000 parts. Individual cavity identification must be provided.

If any defective parts are found during qualification of the cavity, the cavity producing the defective part shall not be used in production. If any defective parts are found when inspection is performed for control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned for corrective action.

All cavities formerly removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test specified above.

4.5.3.2 Stamped Parts.-A sample of at least ten (10) parts shall be dimensionally inspected to qualify the tool used in the stamping process for use in production. In addition, a random sample of five (5) parts shall be selected from the last portion of each hour's production for dimensional inspection as a control of the tool during production.

If any defective parts are found during qualification of the tool, the tool producing the defective part shall not be used in production.

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If any defective parts are found when inspection is performed for control of the tool, the tool producing the defective part shall be removed from production. Further, that portion of production since the last tool control check shall be returned for corrective action.

All tools removed from production because of some fault, may, after reworking, be returned to production providing they pass the qualification test above.

4.5.3.3 Molded Parts.-A sample of five (5) parts (as molded) from each cavity shall be fully inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification. As a control of each cavity during production, the above quantity of parts from each cavity shall be dimensionally inspected after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. If any defective parts are found during qualification of the cavity, the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned for corrective action. All cavities removed from production because of some fault, may, after reworking, be returned to production, providing they pass the qualification test above.

4.5.4 Ballistic Test.-These tests shall be conducted at a Government proving ground in accordance with the applicable Acceptance Test Procedure. The sample cartridges shall be fired in a gun for which the cartridge is standard and observation shall be made for proper functioning and for parts separation or breakup in the gun bore or in flight prior to target impact. Smear cameras shall be positioned to record the initial portion of projectile flight as a minimum.

4.5.4.1 Graze Functioning.-The ammunition for this test shall be conditioned at a temperature of 70 Degrees Fahrenheit (°F) ±5 °F for 16 hours min. prior to firing. Test cartridges shall be fired for graze functioning at a gravel impact field, at a range of 1000 meters.

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5. Preparation for Delivery

5.1 Preservation and Packing (All dimensions are in inches)

5.1.1 Level C.-Package each spoiler switch assembly in a .002 inch thick, Type II, any grade, Finish 1, Spec L-P-378 polyethylene film bag. Size of bag to be 7. open face x 11. deep. The bag may be tapered at the bottom to coincide with the contour of the switch. Switch shall be inserted nose down in the bag, and the excess material at the top inserted within the switch. Pack 30 bagged spoiler switch assemblies in 2 layers of 15 each within a partitioned Type CF, Class Domestic, Variety DW, grade 275, CSSC style fiber box, Spec PPP-B-636. The inside dimensions of the box to be 20 15/16 x 12 9/16 x 14 1/16 high. The partitions shall be of 3 x 5 pattern, and shall slide fit in box. Each cell shall be 4 1/16 square x 6 15/16 high. The material shall be Type CF, Class Domestic, Variety SW, grade 200, "B" flute, Spec PPP-F-320. Place a partition within the box, pack 15 bagged spoiler switch assemblies, nose end up. Place a separator, size 20 1/2 x 12., atop contents and repeat previous step for an additional layer. Separator to be same material as partition. Tape fiber box in accordance with Spec PPP-B-636 as applicable for boxes overpacked or shipped as exterior container.

5.1.2 Level B.-In addition to fiber box specified in Level C, overpack fiber box in a Class E, Spec MIL-B-117 bag after blunting corners of box. Exhaust all excess air within bag and seal bag in accordance with Spec MIL-P-116. Size of bag to be 29. min. open face x 38. min. deep. Pack 18 bagged fiber boxes (2 x 3 x 3 high) in an overseas type, grade B, cleated-plywood box, PPP-B-601, any permissible style. Box to be provided with skids in accordance with Para 3.3.3 of Spec. Requirements for box to be for 350 pound weight of contents (Ref Table III). Inside dimensions of box to be 44 1/2 x 40. x 46. high. Add filler material 39 1/2 x 44. x 1/16 thick on sides, ends and top as required for tight packing. Filler material to comply with Spec MIL-B-3106 or MIL-F-50449.

5.1.3 Level A.-Same as Level B.

5.2 Marking

5.2.1 Level C.-Marking of fiber box to be in accordance with MIL-STD-1169.

5.2.2 Level B.-Marking of fiber box, bag and box to be in accordance with Level A, MIL-STD-1169.

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5.2.3 Level A.-Same as Level B.

6. NOTES

6.1 Intended Use.-The parts covered by this specification are intended for use with the Fuze, PIBD, M509A2 as the fuzing system for Ctg, 105MM, HEAT-T, M456.

6.2 Ordering Data.-See MIL-A-48078 (PA).

6.3 Submission of Designs for Approval.-See 6.2.3 of MIL-A-48078 (PA). Submit equipment designs, as required, to:

Commander
US Army Armament R&D Command
ATTN: DRDAR-QAT-I
Dover, New Jersey 07801

6.4 Inspection Lot Size.-It is noted that the size of inspection lots of components, assemblies or items of delivery may differ from the actual quantities contractually scheduled for delivery. However, in order to facilitate scheduling of test or deliveries, inspection lots of items of delivery may be equivalent to such contract quantities provided all of the lot formation criteria and sampling provisions of this specification are maintained.

6.5 Visual Examination.-When compliance with the applicable requirements is in doubt as a result of visual examination, the characteristic may be measured or gaged to determine acceptability.

6.6 Proving Ground Test Summary.-

TEST	SAMPLE SIZE	REQUIREMENT	TEST PROCEDURES PARAGRAPH
Graze Functioning	20 rds	3.4.1	4.5.4.1
Safety		3.4.2	

Custodian:
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Preparing Activity:
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Project Number: FSC: 1390-A074

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